10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 4.1 — Functions, Composite & Inverse

Unit 4 | Solved Exercise | New Book (PECTAA)

Exercise 4.1 (Solved)

1. If f(x) = 2x + 5 and g(x) = 3x − 2, then find:

(i) f(x) + g(x) (ii) f(x) − g(x)

When we add or subtract two functions, we add or subtract their rules and then simplify by collecting the like terms.

(i) f(x) + g(x)

$f(x) + g(x) = (2x + 5) + (3x - 2)$

$= 2x + 3x + 5 - 2$

$= 5x + 3$

Answer: f(x) + g(x) = 5x + 3

(ii) f(x) − g(x)

$f(x) - g(x) = (2x + 5) - (3x - 2)$

$= 2x + 5 - 3x + 2$

$= 2x - 3x + 5 + 2$

$= -x + 7$

Answer: f(x) − g(x) = −x + 7 = 7 − x

2. If f(x) = x + 2 and g(x) = 2x + 4, then find:

(i) f(x)·g(x) (ii) g(x)·f(x) (iii) f(x)/g(x) (iv) g(x)/f(x)

Multiply the two rules by opening the brackets. In a division, cancel the common factor, and remember the value of x for which the denominator is zero.

(i) f(x)·g(x)

$f(x) \cdot g(x) = (x + 2)(2x + 4)$

$= 2x^2 + 4x + 4x + 8$

$= 2x^2 + 8x + 8$

Answer: f(x)·g(x) = $2x^2 + 8x + 8$

(ii) g(x)·f(x)

$g(x) \cdot f(x) = (2x + 4)(x + 2)$

$= 2x^2 + 4x + 4x + 8$

$= 2x^2 + 8x + 8$

So f(x)·g(x) and g(x)·f(x) are the same

Answer: g(x)·f(x) = $2x^2 + 8x + 8$

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(iii) f(x)/g(x)

$$\frac{f(x)}{g(x)} = \frac{x+2}{2x+4}$$

Take 2 common from the denominator

$$= \frac{x+2}{2(x+2)}$$

Cancel (x + 2), provided $x + 2 \neq 0$, i.e. $x \neq -2$

$$= \frac{1}{2}$$

Answer: f(x)/g(x) = 1/2, $x \neq -2$

(iv) g(x)/f(x)

$$\frac{g(x)}{f(x)} = \frac{2x+4}{x+2}$$

$$= \frac{2(x+2)}{x+2}$$

Cancel (x + 2), provided $x \neq -2$

$= 2$

Answer: g(x)/f(x) = 2, $x \neq -2$

3. For the functions f and g, find (a) (fog)(x) (b) (gof)(x) (c) (fof)(x) (d) (gog)(x).

(fog)(x) means f(g(x)): first put x in g, then put that answer in f. Work from the inside outwards.

Case (i) $f(x) = 2x + 3$, $g(x) = x^3$

(a) $(fog)(x) = f(g(x))$

$= f(x^3)$

$= 2(x^3) + 3$

$= 2x^3 + 3$

(b) $(gof)(x) = g(f(x))$

$= g(2x + 3)$

$= (2x + 3)^3$

$= 8x^3 + 36x^2 + 54x + 27$

(c) $(fof)(x) = f(f(x))$

$= f(2x + 3)$

$= 2(2x + 3) + 3$

$= 4x + 6 + 3$

$= 4x + 9$

(d) $(gog)(x) = g(g(x))$

$= g(x^3)$

$= (x^3)^3$

$= x^9$

Answer: $fog = 2x^3 + 3$, $gof = (2x + 3)^3$, $fof = 4x + 9$, $gog = x^9$

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Case (ii) $f(x) = 2/x,\ x \neq 0,\ g(x) = 2x^2 - 1$

(a) $(fog)(x) = f(g(x))$

$= f(2x^2 - 1)$

$= \frac{2}{2x^2 - 1}$

(b) $(gof)(x) = g(f(x))$

$= g(2/x)$

$= 2(2/x)^2 - 1$

$= 2(4/x^2) - 1$

$= \frac{8}{x^2} - 1$

(c) $(fof)(x) = f(f(x))$

$= f(2/x)$

$= 2 \div (2/x)$

$= 2 \times \frac{x}{2}$

$= x, \quad x \neq 0$

(d) $(gog)(x) = g(g(x))$

$= g(2x^2 - 1)$

$= 2(2x^2 - 1)^2 - 1$

$= 2(4x^4 - 4x^2 + 1) - 1$

$= 8x^4 - 8x^2 + 2 - 1$

$= 8x^4 - 8x^2 + 1$

Answer: $fog = 2/(2x^2 - 1)$, $gof = 8/x^2 - 1$, $fof = x$, $gog = 8x^4 - 8x^2 + 1$

Case (iii) $f(x) = 2x - 1,\ g(x) = (x + 1)/2$

(a) $(fog)(x) = f(g(x))$

$= f(\frac{x + 1}{2})$

$= 2(\frac{x + 1}{2}) - 1$

$= (x + 1) - 1$

$= x$

(b) $(gof)(x) = g(f(x))$

$= g(2x - 1)$

$= \frac{(2x - 1) + 1}{2}$

$= \frac{2x}{2}$

$= x$

(c) $(fof)(x) = f(f(x))$

$= f(2x - 1)$

$= 2(2x - 1) - 1$

$= 4x - 2 - 1$

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$= 4x - 3$

(d) $(gog)(x) = g(g(x))$

$= g\left(\frac{x+1}{2}\right)$

$= \frac{(x+1)/2 + 1}{2}$

$= \frac{x+3}{4}$

Answer: fog = x, gof = x, fof = 4x − 3, gog = (x + 3)/4

4. Find the value of k such that (fog)(x) = (gof)(x), where f(x) = 3x + 2 and g(x) = 6x − k.

Solution

First find $(fog)(x) = f(g(x))$

$= f(6x - k)$

$= 3(6x - k) + 2$

$= 18x - 3k + 2$

Now find $(gof)(x) = g(f(x))$

$= g(3x + 2)$

$= 6(3x + 2) - k$

$= 18x + 12 - k$

Given that $(fog)(x) = (gof)(x)$

$18x - 3k + 2 = 18x + 12 - k$

Cancel 18x from both sides

$-3k + 2 = 12 - k$

$-3k + k = 12 - 2$

$-2k = 10$

$k = -5$

Answer: k = −5

5. Given that f(x) = 3x + 2 and g(x) = 2x + 3, find:

(i) g(f(4)) (ii) f(f(3)) (iii) f(g(−2))

(i) g(f(4))

First $f(4) = 3(4) + 2$

$= 12 + 2$

$= 14$

Now $g(14) = 2(14) + 3$

$= 28 + 3$

$= 31$

Answer: g(f(4)) = 31

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(ii) f(f(3))

$f(3) = 3(3) + 2$

$= 9 + 2$

$= 11$

Now $f(11) = 3(11) + 2$

$= 33 + 2$

$= 35$

Answer: f(f(3)) = 35

(iii) f(g(−2))

$g(-2) = 2(-2) + 3$

$= -4 + 3$

$= -1$

Now $f(-1) = 3(-1) + 2$

$= -3 + 2$

$= -1$

Answer: f(g(−2)) = −1

6. Find $f^{-1}(x)$ in each of the following:

Method: write y = f(x), make x the subject, then replace y by x. The result is $f^{-1}(x)$.

(i) f(x) = 2x − 3

Let $y = 2x - 3$

$2x = y + 3$

$x = \frac{y + 3}{2}$

Replace y by x

$f^{-1}(x) = \frac{x + 3}{2}$

Answer: $f^{-1}(x) = (x + 3)/2$

(ii) $f(x) = 4x^3 - 1$

Let $y = 4x^3 - 1$

$4x^3 = y + 1$

$x^3 = \frac{y + 1}{4}$

Take the cube root of both sides

$x = \sqrt[3]{((y + 1)/4)}$

Replace y by x

$f^{-1}(x) = \sqrt[3]{((x + 1)/4)}$

This may also be written as $((x + 1)/4)^{1/3}$

Answer: $f^{-1}(x)$ = ((x + 1)/4) raised to the power 1/3

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(iii) $f(x) = \sqrt{(x - 1)}, x \geq 1$

Let $y = \sqrt{(x - 1)}$

Squaring both sides

$y^2 = x - 1$

$x = y^2 + 1$

Replace y by x

$f^{-1}(x) = x^2 + 1$

Since the range of f is $y \geq 0$, the domain of f^{-1} is $x \geq 0$

Answer: $f^{-1}(x) = x^2 + 1, x \geq 0$

(iv) $f(x) = (x + 1)/(3x - 2), x \neq 2/3$

Let $y = \frac{x + 1}{3x - 2}$

$y(3x - 2) = x + 1$

$3xy - 2y = x + 1$

$3xy - x = 2y + 1$

$x(3y - 1) = 2y + 1$

$x = \frac{2y + 1}{3y - 1}$

Replace y by x

$f^{-1}(x) = \frac{2x + 1}{3x - 1}$

Answer: $f^{-1}(x) = (2x + 1)/(3x - 1), x \neq 1/3$

7. The functions f and g are defined such that $f(x) = 4x + 2$ and $g(x) = 6x - 18$.

(i) Find $f^{-1}(x)$ and $g^{-1}(x)$.

(ii) Find x if $f^{-1}(x) = g^{-1}(x)$.

(i) $f^{-1}(x)$ and $g^{-1}(x)$

Let $y = 4x + 2$

$4x = y - 2$

$x = \frac{y - 2}{4}$

$f^{-1}(x) = \frac{x - 2}{4}$

Now let $y = 6x - 18$

$6x = y + 18$

$x = \frac{y + 18}{6}$

$g^{-1}(x) = \frac{x + 18}{6}$

Answer: $f^{-1}(x) = (x - 2)/4$, $g^{-1}(x) = (x + 18)/6$

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(ii) x if $f^{-1}(x) = g^{-1}(x)$

$\frac{x-2}{4} = \frac{x+18}{6}$

Cross multiply

$6(x-2) = 4(x+18)$

$6x - 12 = 4x + 72$

$6x - 4x = 72 + 12$

$2x = 84$

$x = 42$

Answer: x = 42

8. Verify that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$

Put $f^{-1}(x)$ inside f and simplify. The answer must come out x both ways, which proves that the two functions undo each other.

(i) f(x) = x − 6

First find $f^{-1}(x)$: let $y = x - 6$, so x

$= y + 6$

$f^{-1}(x) = x + 6$

$f(f^{-1}(x)) = (x + 6) - 6$

$= x$

$f^{-1}(f(x)) = (x - 6) + 6$

$= x$

Answer: $f(f^{-1}(x)) = f^{-1}(f(x)) = x$ (verified)

(ii) f(x) = 7x − 4

Let $y = 7x - 4$, so $7x$

$= y + 4$

$f^{-1}(x) = \frac{x+4}{7}$

$f(f^{-1}(x)) = 7(\frac{x+4}{7}) - 4$

$= (x + 4) - 4$

$= x$

$f^{-1}(f(x)) = \frac{(7x-4)+4}{7}$

$= \frac{7x}{7}$

$= x$

Answer: $f(f^{-1}(x)) = f^{-1}(f(x)) = x$ (verified)

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(iii) f(x) = (x − 3)/4

Let $y = \frac{x-3}{4}$, so $4y = x - 3$

$x = 4y + 3$

$f^{-1}(x) = 4x + 3$

$f(f^{-1}(x)) = ((4x + 3) - 3)/4$

$= \frac{4x}{4}$

$= x$

$f^{-1}(f(x)) = 4(\frac{x-3}{4}) + 3$

$= (x - 3) + 3$

$= x$

Answer: $f(f^{-1}(x)) = f^{-1}(f(x)) = x$ (verified)

(iv) f(x) = (x − 4)/(x + 2), x ≠ −2

Let $y = \frac{x-4}{x+2}$

$y(x + 2) = x - 4$

$xy + 2y = x - 4$

$xy - x = -4 - 2y$

$x(y - 1) = -4 - 2y$

$x = \frac{-4-2y}{y-1}$

$f^{-1}(y) = \frac{-4-2y}{y-1}$

Replace y by x at both places

$f^{-1}(x) = \frac{2x+4}{1-x}$

Now verify. Put $f^{-1}(x)$ in place of x in f

$f(f^{-1}(x)) = f((2x + 4)/(1 - x))$

Multiply the numerator and the denominator by $(1 - x)$

$= \frac{(2x+4)-4(1-x)}{(2x+4)+2(1-x)}$

$= \frac{2x+4-4+4x}{2x+4+2-2x}$

$= \frac{6x}{6}$

$= x$

In the same way $f^{-1}(f(x))$ also simplifies to x

Answer: $f(f^{-1}(x)) = f^{-1}(f(x)) = x$ (verified)

9. Without finding $f^{-1}(x)$, find domain and range of $f^{-1}(x)$:

(i) $f(x) = 12x - 3$ (ii) $f(x) = ½x + 8$

The domain of f^{-1} is the range of f, and the range of f^{-1} is the domain of f. A linear function with no restriction has domain and range equal to all real numbers.

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(i) f(x) = 12x − 3

f is a linear function, so it is defined for every real x

Domain of f = R, and its range is also R

Therefore domain of f^{-1} = R

And range of f^{-1} = R

Answer: Domain = R, Range = R

(ii) f(x) = ½x + 8

f is linear, so it is defined for every real x

Domain of f = R, and its range is also R

Therefore domain of f^{-1} = R

And range of f^{-1} = R

Answer: Domain = R, Range = R

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